INTEROFFICE MEMORANDUM

To: CIO

From: Team 02

Date: October 24, 2026

Subject: High Availability Solution Design

Summary:

Below is a plan outlining the implementation of a second Palo Alto firewall utilizing HA links to ensure no single point of failure. This plan also covers the implementation of a second internet feed provider.

Existing Hardware

Currently, our network topology supports a single Palo Alto PA-VM, which acts as a single source of failure for the firewall, which connects and filters the packets being sent from the External Network Segment (our VyOS Router) to the internal network segments.

Proposed Hardware and Configuration

To remediate this, Security Team recommends the purchase and implementation of second Palo Alto PA-VM Firewall. This device will work with the existing Palo Alto Firewall to ensure a seamless and transparent failover. This firewall should match the existing firewall's model, PAN-OS version, and CPU allocation.

The secondary Palo Alto Firewall will act as a passive Standby, utilizing a "heartbeat" ping message to ensure the health of the active firewall and a transition should the active firewall go down. The firewalls should be connected utilizing Palo Alto's available High-Availability (HA) links. HA1 will act as the Control Link to exchange heartbeats, mange control-plane communication, and ensure device health. HA2 will act as the Data Link, which will manage data-plane and session state synchronization.

To connect these devices, Team 02 recommends utilizing VLANs since the firewall connections are also virtual. To ensure no single point-of-failure, it is essential that the second firewall is hosted on a separate host. Ensure that appropriate anti-affinity rules are set.

Wiring Setup

The following diagram shows the wiring topology to interconnect the firewalls using Cat6 Ethernet Cables. Source: https://storage.ghost.io/c/27/3a/273a3b8f-882f-4589-9659-f17410001182/content/images/2022/04/diagram-with-ip-v1.PNG

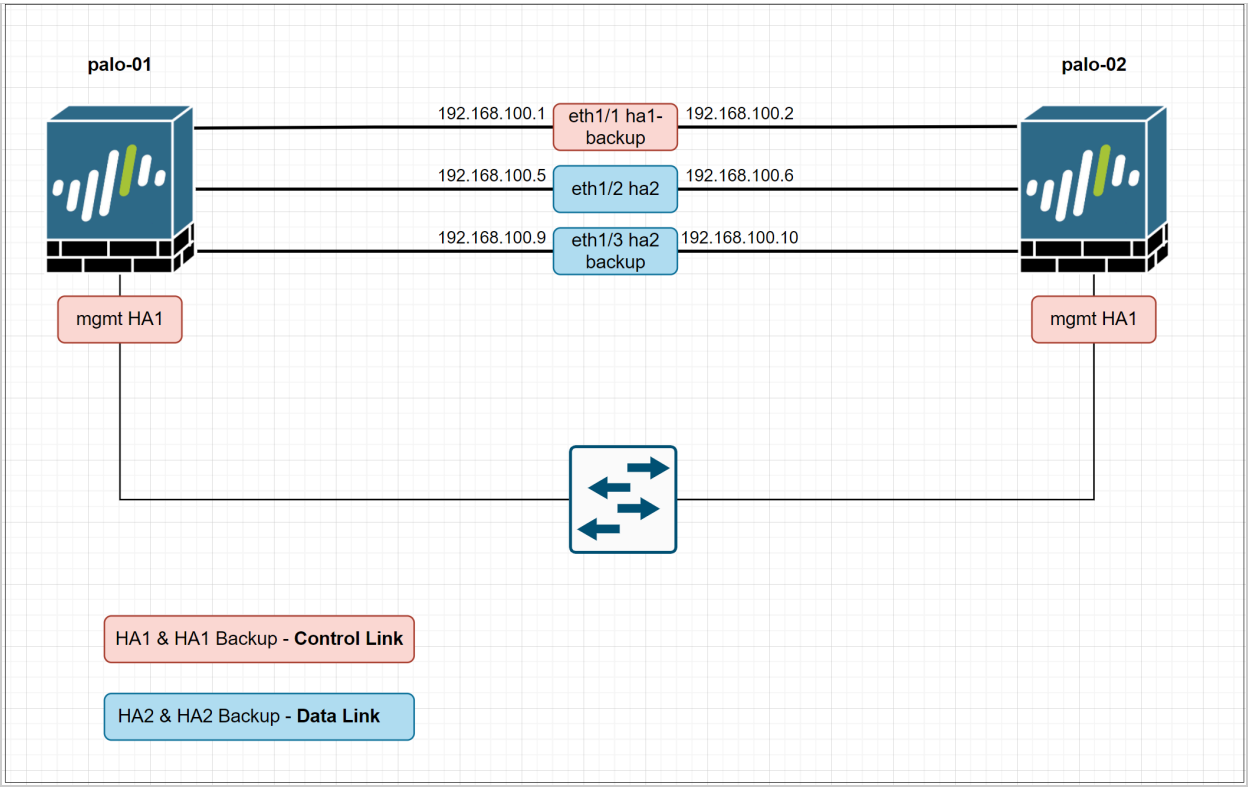


Figure 1. Palo Alto firewall connection topology

Affect of Second Internet Service Provider

If a second internet feed provider was included, this would be wired into the VyOS router to ensure proper failover. The VyOS router can be configured for automatic WAN failover which will ensure that a failure in one provider will automatically trigger a switch to the second provider. Ensure that a second ISP circuit is added, along with its provider handoff device. The HA configuration of the firewalls would remain the same.

References

Official Palo Alto HA documentation can be found here: https://docs.paloaltonetworks.com/ngfw/administration/high-availability/ha-ports-on-palo-alto-networks-firewalls.

If there are any concerns, questions, or clarifications, please do not hesitate to reach out.

With Regards,
Team 02